

Bankruptcy Cascades in Interbank Markets

An Agent-Based Model of Systemic Risk

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- Study of credit networks and interbank systems using an **Agent-Based Model (ABM)** [1].
- Analysis of the relationship between **business cycles** and bankruptcy cascades [1].
- Modeling of a three-sector economy: goods, credit, and interbank market [1].
- Identification of the trade-off between risk sharing and systemic risk [1].

Motivation: Financialization of the Economy

- In the last 30 years, the financial sector has outgrown the production sector [2].
- Shift from loan-based financing to speculative, market-based activities [3].
- Increased bank leverage and corporate debt-equity ratios have made the economy fragile [3, 4].
- Mainstream theory (Arrow-Debreu) fails to explain complex agent interactions during crises [5].

Propagation of Systematic Failure

The literature identifies three primary types of contagion:

- 1 **Bank runs:** Known as self-fulfilling panics [6].
- 2 **Asset price contagion:** Spirals caused by fire sales [6].
- 3 **Inter-locking exposures:** Direct linkages through the interbank market [6].

This paper focuses on the interbank market as a mechanism for liquidity crisis contagion [6].

The Trade-off: Connectivity vs. Risk

- **Benefit:** Increasing connectivity allows for **risk sharing** and diversification [7, 8].
- **Cost:** Excessive connectivity increases **systemic risk** by acting as a vector for contagion [8, 9].
- Evidence suggests a **non-monotonic relationship** between connectivity and systemic risk in the presence of agent heterogeneity [9, 10].

Structure of the Model

The model consists of three interacting sectors:

- 1 **Goods Market:** Demand-driven output with incomplete information [11].
- 2 **Credit Market:** Firms borrow from banks to finance investment [12].
- 3 **Interbank Market:** Banks share risk and liquidity through random connectivity [13, 14].

POPULATION: F_t firms and B_t banks [11].

The Goods Market

- Output is demand-driven; firms sell as much as the market can absorb [11].
- Incomplete information creates a gap between expected demand $E(D_{f,t})$ and realized demand $D_{f,t}$ [11, 15].
- This gap can generate unexpected shocks to revenues and negative profits [12].

Firms: Production and Capital

- Production function: $Y_{f,t} = \phi K_{f,t}$, where ϕ is constant capital productivity [16].
- Decisions follow **boundedly rational heuristics** under uncertainty [15].
- Firms produce based on forecasted demand; if demand exceeds capacity, they seek loans to increase capital [17, 18].

- Loan demand to reach expected demand:

$$L_{f,t} = \max \left\{ \frac{E(D_{f,t})}{\phi} - Y_{f,t}, 0 \right\} \quad [18]$$

- Adjusted for risk aversion (α) and financial fragility:

$$L_{f,t}^d = \alpha \left(1 - \frac{\Lambda L_{f,t}}{E(\pi_{f,t+1})} \right) L_{f,t} \quad [19]$$

- Firms decrease loan demand if they expect profits won't cover installments [19, 20].

- Real terms profit: $\pi_{f,t} = \min(D_{f,t}, Y_{f,t}) - \Lambda L_{f,t}$ [20].
- All profits are retained as capital [21].
- Capital stock evolution:

$$K_{f,t} = K_{f,t-1} + \pi_{f,t} + L_{f,t}^d \quad [21]$$

Banks: Interest Rate Policy

- Firms contact a small number of random banks [21].
- Interest rate offered by bank b :

$$i_{f,b}^t = \bar{i} + c \left(\frac{L_{f,t}^d}{S_{b,t}} \right)^a \quad [22]$$

- $\bar{i}$ is set by the Central Bank, $S_{b,t}$ is the bank's liquidity [22].
- Rates decrease as the bank's **financial robustness** increases [22].

- **Basel I/II:** Regulation forces banks to hold capital caution [23].
- Modeled as a probability γ of granting a loan:

$$\gamma_{f,b}^t = 1 - b \left(\frac{L_{f,t}^d}{S_{b,t}} \right)^s \quad [23]$$

- b is a regulatory parameter; increasing b forces higher reserves and reduces risk [23, 24].

Interbank Market Mechanism

- If a bank lacks liquidity to cover a firm's loan, it enters the interbank market [25].
- It asks for funds from x randomly chosen banks [25].
- Connectivity is modeled as a **random graph** [13].
- Borrower banks choose the lender offering the lowest interest rate [26].

Bank Liquidity Evolution

Liquidity $S_{b,t}$ evolves according to:

$$S_{b,t} = S_{b,t-1} - \sum L_{f,t}^d + \frac{1}{\tau} \sum L_{f,t'}^d (1 + i_{f,b}^{t'}) + I_{b,t} \quad [27]$$

- Second term: Total loans granted [27].
- Third term: Installments and interest from "safe" firms [27].
- Fourth term ($I_{b,t}$): Interbank borrowing or lending [27].

Profit $\pi_{b,t}$ depends on:

- ① Interest paid by firms [28].
- ② **Firms' bad debt:** Losses from bankrupt firms (ν share) [28, 29].
- ③ Interbank credit interactions [29].

Bankruptcy occurs if the agent lacks enough cash to repay loans [30].

Bankruptcy and Entry Processes

- Bankrupt agents leave the market [30].
- New entrants N_t^{entry} are determined by expected profit opportunities, linked to the average interest rate [31, 32].
- Entrants are on average smaller than incumbents, mimicking empirical literature [33, 34].

The model successfully reproduces:

- **Endogenous growth** with persistent fluctuations [35].
- **Volatility clustering:** Large changes in variables tend to cluster together [36].
- Autocorrelation of absolute growth rates (≈ 0.95), matching G7 empirical data (0.93) [36, 37].

- Recessions are triggered by the failure of **large firms** [38].
- Disequilibrium dynamics lead to self-organization with persistent excess demand [38].
- Crises depend not on the quantity of bankruptcies, but on the **quality** (size) of bankrupted agents [39].

Minsky's Financial Instability Hypothesis

- Prosperity leads to optimism; banks grant more loans and underestimate risk [40, 41].
- Positive correlation (≈ 0.63) between output and granted loans [41].
- Rapid increases in **financial fragility** (high leverage) anticipate severe crises [42].

Emergent Heterogeneity

- Trading generates **fat tail distributions** for firm and bank sizes [43].
- The economy is dominated by small/medium units, while a few large units control a significant part of supply [43].
- This makes the "representative firm" notion meaningless [44].

The Role of Reserve Requirements

- Increasing the reserve ratio b decreases bank failure rates [45].
- **Individual stability:** Lower average bank leverage [45].
- **Macroeconomic cost:** Reduces output growth by decreasing firm investment and granted loans [45, 46].

Interbank Linkages (x)

- As interbank connectivity (x) increases, more banks fail in each period [47].
- Highly connected systems exhibit **rapid survival curve declines** [48, 49].
- Connectivity acts as a vector for "contagious failures" [49].
- Higher connectivity has no effect on real output growth before cascades occur [50].

Large Bankruptcy Cascades

- Measurement of the largest connected component of failed banks [51].
- A highly interconnected market leads to **larger cascades** of bankruptcies [51, 52].
- The systemic risk from connectivity outweighs the insurance benefits of risk sharing [53].

Kurtosis and Hill Exponents

To measure the probability of extreme events:

- **Kurtosis:** Increases with connectivity, indicating a higher probability of systemic collapse [52].
- **Hill Exponent:** Decreases from "normal" values (4) toward "fragile" values (2) as interbank links increase [54].
- This proves that high connectivity generates **fatter tails** in failure distributions [54].

Conclusions and Policy

- **Heterogeneity** is a root cause of systemic instability; failure of large banks is catastrophic [55, 56].
- **Policy Insight:** Interbank lending should ideally be restricted among banks with similar liquidity characteristics [57].
- **Redundancy:** Reserve requirements dampen cascades but limit growth [55].

- The model provides a tool to understand the 2008 financial meltdown [1].
- Confirms the **procyclicality of leverage** [58].
- Highlights that risk diversification across many agents has **ambiguous** effects on total systemic stability [59].